MIL-P-37083A
15 June 1987
SUPERSEDING
MIL-P-37083
9 April 1975 and
MIL-P-37361
4 February 1977

MILITARY SPECIFICATION

PLIERS, DENTAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers dental pliers suitable for holding and placing pins and points.

1.2 Classification. Dental pliers covered by this specification shall be of the following numbers, (see 6.1):

No. 102
No. 111
No. 137

2. APPLICABLE DOCUMENTS

2.1 Government documents.

AMSC N/A FSC 6520

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-RST, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Federal

GG-I-526	Instruments, Dental and Surgical: General Specifications.
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-601	Boxes, Wood, Cleated-plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	Boxes, Fiberboard.
PPP-B-676	Boxes, Setup.

Military

MIL-P-116	Methods of Preservation.
MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Federal

FED-STD-151	Metal, Test Methods.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-147	Palletized Unit Loads.
MIL-STD-794	Parts and Equipment, Procedures for Packaging of.

DRAWINGS

Defense Personnel Support Center (DPSC)

2179	Pliers, Dental, No. 102
20760	Pliers, Dental, No. 111
20500	Pliers, Dental, No. 137

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(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issue of documents not listed in the DoDISS shall be the issue of the nongovernment document which is current on the date of the solicitation.

American Society for Testing and Materials (ASTM)

ASTM E 18 - Standard Test Methods for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.

ASTM E 92 - Standard Test Method for Vickers Hardness of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General requirements. The pliers shall be in accordance with the applicable requirements of GG-I-526 with the options and additional requirements specified herein.

3.2 Material. Pliers shall be fabricated from either carbon steel or stainless steel as specified in table I. At the time of proposal, the offeror shall specify which material option is to be furnished, see 6.1.

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3.3. Hardness. After proper stress relieving, the pliers shall have the Rockwell hardness as specified in table I for the applicable material. Hardness shall be determined as specified in 4.4.2.

3.4 Distortion. The handle of the pliers shall not show pliability or distortion under normal usage.

TABLE I. Material, hardness and plating.

Material		Rockwell Hardness	Plating
Carbon	Carbon content: 0.40 to 1.10 present Manganese content: 0.45 to 0.90 per - cent except when carbon cont is over 0.90 percent, the manganese con- tent shall be a max- imum of 0.50 per- cent. Phosphorous content: Maximum of 0.040 percent. Sulfur content: Maximum of 0.050 percent.	C40 to C45, with exception of the screw or pin. When the carbon content is over 0.90 percent, the hardness shall be C48 to C52.	Plating shall be chromium over nickel plated in accordance with GG-I-526 except the minimum tickness shall not apply, provided the pliers comply with the boil test.
Stain- less Steel	Class 4, Type 420	C40 to C48, with exception of the screw or pin.	None
	Class 4, Type 410	C40 to C45, with the exception of of the screw or pin.	None

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3.5 Style, design and dimensions. The pliers shall be suitable for thepurpose specified in table II and shall conform to the Defense Personnel Support Center (DPSC) drawings specified in table II for the applicable number. The pliers shall be furnished in the number specified. Number 102 shall be a cone socket pliers. Number 111 shall be a crown pliers, with serrated beaks. Number 137 shall be a contouring pliers.

Table II. Drawing Number and Purpose of each Instrument

Number	DPSC Drawings	Purpose
No. 102	2179	Holding, tightening and removal of cone socket points (in and from handles), and general laboratory use.
No. 111	20760	Ligating, placing and removing pins.
No. 137	20500	General contouring work in the dental laboratory.

3.5.1 Drawings. The drawings forming a part of this specification are specification control drawings. The supplier shall be responsible for preparing his own shop drawings. Where tolerances prescribed could cumulatively result in improper design, the supplier shall provide a product with dimensions within the tolerances prescribed on the drawing to insure proper function, fit, assembly or operation of the item. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer.

3.6 Locks. The joint of the pliers shall be box-lock or lap-joint type. The edges of joints shall be properly beveled to prevent pinching of or injury to tissue.

3.7 Finish. In addition to the requirements of GG-I-526, the pliers shall have a mirror, satin, matte or a smooth, non-reflective surface finish. The inside face of the jaws of the pliers No. 102 and No. 111 shall be serrated as specified in the applicable drawing, the inside face of the jaws of the pliers No. 137 jaws shall be smoothly contoured surfaces as specified in the applicable drawing.

3.8 Plating. Carbon steel pliers shall be plated in accordance with the requirements specified in Table I.

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3.9 Corrosion resistance. All pliers shall comply with the boil test requirements of GG-I-526 when tested as specified in 4.4.3.1. Pliers fabricated from stainless steel shall also meet the copper sulfate test requirements of GG-I-526 when tested as specified in 4.4.3.2.

3.10 Identification marking. The pliers shall be marked with the applicable instrument number. Marking shall be permanent and legible. Stainless steel pliers of foreign manufacture shall also be marked with the word "Stainless".

3.11 Workmanship. The pliers shall be free from defects which may detract from their appearance or impair their serviceability, proper functioning, or intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

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4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in table III and V (see 4.2.1 and 4.3) and table IV (4.2.2).

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQLs) shall be as indicated in table III. Unit of product for examination purposes shall be one pliers.

Table III. Inspection and acceptable quality levels

Type defect	Inspection level	AQL (percent defective)
Major A	II	1.0
Major B	S-2	1.5
Minor	II	2.5

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in table IV. Acceptance number shall be zero for all sample sizes. Lots for hardness testing shall be as specified in 4.2.2.2. The unit of product for testing purposes shall be one pliers. Where feasible, the same unit of product shall be used for the determination of two or more test characteristics.

Table IV. Sampling criteria for tests

Characteristic	Requirement	Test Procedures	Inspection Level
Material	3.2	4.4.1	4.2.2.1
Hardness	3.3	4.4.2	S-2
Boil Test	3.9	4.4.3.1	S-2
Copper sulfate	3.9	4.4.3.2	S-2

4.2.2.1 Sampling of raw material. A sufficient quantity from each lot of metal used in the construction of the pliers shall be used for composition analysis.

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4.3 Examination. The pliers shall be examined to determine compliance with requirements contained in this specification. Examination shall be conducted in accordance with the following classification of defects. Examination shall include, but shall not be limited to, the defects listed in table V.

Table V. Classification of defects

Categories	Defects
Major A:	
101	Number and design not as specified.
102	Finish not as specified.
103	Plating of plated pliers not free from flaking or blisters.
104	Plating of plated pliers incomplete, porous, or not continuous.
105	Jaws do not align properly.
106	Joint of pliers not free from stiffness.
107	Joint of pliers has excessive play.
108	Edges of the joint of pliers sharp (not properly beveled).
109	Pliers not free from cracks.
110	Inside faces of beaks not serrated, (No. 102 and No. 111).
Major B:	
151	Dimensions not as specified.
Minor:	
201	Identification marking not present, legible, or complete.
202	Pliers not permanently marked with the name or trademark of the contractor.

4.4 Tests. Tests shall be conducted to determine compliance with the requirements of this specification (see 4.2.2.).

4.4.1 Chemical composition. In the absence of certificates of quality for the raw material, chemical analysis shall be conducted in accordance with method 111 or 112 or FED-STD-151. In case of dispute, method 111 shall govern.

4.4.2 Hardness. Hardness shall be determined in accordance with ASTM Designation E18 (Rockwell) or E92 (Vickers or Diamond Pyramid).

4.4.3 Corrosion resistance.

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4.4.3.1 Boil test. The pliers shall be subjected to the boil test, as specified in GG-I-526.

4.4.3.2 Copper sulfate test. The pliers shall be subjected to the copper sulfate test, as specified in GG-I-526.

4.5 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. Pliers shall be cleaned and dried in accordance with MIL-P-116.

5.1.1.2 Preservative application. Preservatives shall not be used.

5.1.1.3 Unit package. Each pair of pliers shall be packaged in accordance with MIL-P-116, method IC-1. Closure shall be as specified in the applicable box specification.

5.1.1.4 Intermediate package. Six unit packages containing pliers of one number only shall be packaged in a box conforming to PPP-B-566, PPP-B-636, class domestic or PPP-B-676. Closure shall be as specified in the applicable box specification.

5.1.2 Level C. The level C preservation for pliers shall conform to the MIL-STD-794 requirements for this level.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.2.1 Level A. Twenty-four intermediate packages (144 units) containing pliers of one number only shall be packed in an exterior container desgined for a type 2 load and conforming to PPP-B-601, overseas type or PPP-B-621, class 2. Closure and strapping shall be as specified in the applicable box specification.

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5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall be as specified in the liner specification.

5.2.2 Level B. Twenty-four intermediate packages (144 units) containing pliers of one number only shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, domestic type, PPP-B-621, class 1 or PPP-B-636, class domestic. Closure shall be as specified in the applicable box specification.

5.2.3 Level C. The level C packing for pliers shall conform to the MIL-STD-794 requirements for this level.

5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.2.5 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination exceed 250 pounds or 20 cubic feet. Loads shall be unitized on pallets conforming to MIL-STD-147. Quantities less than 20 cubic feet need not be unitized. Pallet loads including the pallet shall not exceed 54 inches in height, 48 inches in width, and 40 inches in length. Pallet shall conform to Type IV or Type V.

5.2.5.1 Levels A and B. Pliers of one number only packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets conforming to MIL-STD-147.

5.2.5.2 Level C. Pliers packed as specified in 5.2.3 shall be unitized as specified in 5.2.5.

5.3 Marking.

5.3.1 Levels A, B, and C. Each unit pack, intermediate package, exterior container and unitized load shall be marked as specified in MIL-STD-129.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

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5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.5.

6. NOTES

6.1 Ordering data.

6.1.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Number required (see 1.2).
c. Material option (see 3.2).
d. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 National Stock Number (NSN) coverage. This specification covers the following items listed in the Federal Supply Catalog.

National Stock Number	Item Identification
6520-00-542-7480	PLIERS, DENTAL, No. 102, 5-7/6 inch (13.81 cm)
6520-00-543-0150	PLIERS, DENTAL, No. 111, 5-1/8 inch (13.01 cm)
6520-00-543-3750	PLIERS, DENTAL, No. 137, 5-1/4 inch (13.33 cm)

6.3 Subject term (key word) listing.
Dental
Pliers
Holding
Contouring

MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force-03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

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Review Activities:

Army-MD
Navy-MS
Air Force-03

Project No. 6520-1922

Location: NCR Disk #7

CIVIL AGENCY COORDINATING ACTIVITIES:
VA-OSS
PHS
FDA-MPQAS

05756
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-37083A	PLIERS, DENTAL

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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